

Supplementary Figure S1. Representative Model-Route Schematics

Reviewer-oriented structural guide to the current manuscript line

Drug-Only

Exp6

Drug: Morgan fingerprint (2048)

Encoder: compact MLP

Enzyme branch: none

Structure-only lower bound.

Morgan-CNN

Exp1

Drug: Morgan fingerprint (2048)

Enzyme: tokenized sequence

Fusion: sequence CNN + compact head

Strong released-split lightweight baseline.

Lightweight Enzyme-ID Recovery

Exp7-plus

Drug: Morgan + 8 descriptors

Enzyme: ID embedding only

Fusion: 160-dim joint classifier

Minimal enzyme ID recovers much of the signal.

Two-Tower V1

Exp2

Drug tower: 2056 -> 32

Enzyme tower: sequence CNN + ID

Predictor: 64-dim joint MLP

Reference explicit dual-branch route.

CNN-2 Targeted Refinement

Exp13

Drug tower: fixed Exp13 route

Enzyme branch: 2-layer CNN

Scope: targeted enzyme-side depth change

Focused enzyme-side refinement over broad expansion.

Exp8-11 remain localized modifications on the Two-Tower V1 line, so they are summarized textually rather than redrawn as separate full architectures.